ANGU PRAVEEN MUTHU SWAMY

Computational Engineer | FEA | Thermal | Simulation | Predictive Maintenance AI

MOBILE	+49-15115855584
EMAIL	angupraveenms@gmail.com
LINKEDIN	linkedin.com/in/angu-praveen/
LOCATION	Braunschweig, Germany
PORTFOLIO	angupraveenms.github.io/portfolio/
VISA	Permanent Resident

PROFILE

Computational Engineer with 4+ years of experience in Finite Element Analysis (FEA), Computational Fluid Dynamics (CFD), thermal simulation, and Multiphysics Simulation across semiconductor, automotive, and aerospace applications. Specialized in Thermal Management, Structural Analysis, Design Optimization, and model order reduction. Experienced in developing automated simulation workflows and AI-driven predictive maintenance solutions using Python and MATLAB.

ABAQUS / ANSYS	100%
PYTHON / MATLAB	100%
SU2 / OPENFOAM	90%
AI & MACHINE LEARNING	80%
WEB DEV (REACT / JS)	70%
C / C++	70%

Windows Linux MacOS GitHub SVN

EDUCATION

10/2017 - 06/2021 **Master of Science, M.Sc** — Technische Universität Braunschweig, Germany
Computational Sciences in Engineering — **Grade: 1.9**

08/2012 - 08/2016 **Bachelor of Engineering, B.E.** — Dayananda Sagar College of Engineering, India
Mechanical Engineering — **Grade: 81%**

PROFESSIONAL EXPERIENCE

09/2025 - Present **Professional Doctorate Candidate** — Amsterdam University of Applied Sciences, Amsterdam

- Developing an explainable Predictive Maintenance prototype and Digital Twin frameworks using legacy shipboard data to deliver transparent maintenance decisions for marine auxiliary systems.
- Engineering interpretable AI models (GAM, EBM) tailored for Condition Monitoring, integrating explainability metrics directly into operator workflows for verifiable insights.
- Designing a decision-support dashboard to visualize health indices, anomaly indicators, and Remaining Useful Life (RUL) during vessel field trials.

01/2022 - 01/2025 **Computational Engineer** — Capgemini Engineering GmbH, Munich

- Executed 3D thermo-mechanical simulations for EUV lithography systems to evaluate structural integrity and thermal behavior.
- Developed steady-state and transient thermal models using ANSYS, accelerating solver convergence by 20%.
- Optimized cooling circuit designs, driving a 16% increase in overall thermal management efficiency.
- Conducted comprehensive FEM result evaluations, sensitivity studies, and hotspot analyses to maximize component lifetime.
- Applied Model Order Reduction (MOR) methodologies to decrease matrix size by 96% while preserving 10^{-8} numerical accuracy.
- Engineered MATLAB and VBA automation frameworks to streamline encoder testing, achieving up to 40% processing time reduction and 90% fewer manual errors.

11/2020 - 06/2021 **Master Thesis** Robert Bosch Automotive Steering GmbH, Schwäbisch Gmünd

- Developed an automated strength verification framework for Electronic Power Steering bolts using ABAQUS and Python, reducing analysis time by **90%**.
- Implemented VDI 2230-compliant joint analysis with $<$**0.5% deviation** from physical tests through Python-based automation.
- Built custom ABAQUS plugins and parametric bolt models enabling standardized and repeatable joint analysis.
- Achieved **3**$\times$ **faster** design iterations by eliminating manual Excel-based workflows.
- Enhanced computational efficiency by **40%** while improving result reliability through experimental validation.

09/2018 - 10/2020 **Scientific Assistant** Open Hybrid Lab Factory GmbH, Wolfsburg

- Performed advanced ANSYS simulations on FRP and hybrid materials, achieving higher accuracy in stress predictions and a **25%** thermal stress reduction through design optimization.
- Engineered crack detection and prediction workflows using ImageJ and Python-based machine learning algorithms, achieving over **90%** prediction accuracy.
- Created VBA-based automation tools reducing data processing time by **50%** across research team workflows.
- Programmed **3 custom UMAT subroutines** for ABAQUS enabling advanced nonlinear material analysis of potting materials.

KEY PROJECTS

2019 - 2020 **Aerodynamic Optimization Project** Technische Universität Braunschweig, Germany

- Executed multi-fidelity design optimization for a long-range Blended Wing Body aircraft.
- Engineered a SUAVE-based workflow integrating OpenVSP for geometry, GMSH for meshing, and SU2 for high-fidelity computational fluid dynamics.
- Implemented performance estimation for fuel efficiency and maximum takeoff weight using SUAVE's Python-based aircraft sizing algorithms.
- Slashed total aerodynamic drag by 75.95% (from 0.018629 to 0.004480) through optimized pressure distribution and Free Form Deformation (FFD).
- Achieved a 38.96% reduction in Maximum Takeoff Weight (MTOW) while boosting overall fuel efficiency by 35.96%.

CORE COMPETENCIES & TECHNOLOGIES

ANSYS **ABAQUS** **OpenFOAM** **SU2** **SUAVE** **OpenVSP** **ICEM CFD** **GMSH**
Python **MATLAB** **C/C++** **VBA** **APDL** **SQL** **Explainable AI (GAM/EBM)**
Deep Learning (ANN) **Physics-Informed ML** **scikit-learn** **ImageJ** **Digital Twin**
Condition Monitoring **Git** **GitHub** **SVN** **Linux** **macOS** **Windows**

LANGUAGES

English - Fluent (IELTS 7.5)
German - Proficient (DSH 2/C1)
Tamil - Mother Tongue
Hindi, Kannada - Fluent

INTERESTS

Swimming
Adventure Sports
Hiking
Motorcycling

CERTIFICATIONS

ANSYS APDL Programming
Deep Learning A-Z™: Hands-On ANN
Complete Modern C++ (C++11/14/17)
Machine Learning by Andrew NG

Angu Praveen MS Bengaluru, April 24, 2026